Linear Algebra — Mid-Term Model Test 2

Covers Sections 1.1–1.3 and 2.1–2.6 · Level: moderate

Instructions. Time allowed: 120 minutes. Calculators are *not* allowed. Show all working. Total: 100 points. Difficulty: moderate (a step above the sample test).

1 Vectors

Let $v = (1, 2, 1)$, $w = (2, 1, -1)$.

(a) Is $b = (4, 5, 1)$ a linear combination $cv + dw$? If so, find c and d. **[7 pts]**

(b) Find the angle between v and w. **[6 pts]**

(c) Suppose $\|x\| = 3$ and $\|y\| = 4$. What are the largest and smallest possible values of $\|x - y\|$? What is $\|x - y\|$ if $x \perp y$? **[7 pts]**

2 Matrix operations

(1) Calculate the multiplication **[8 pts]**

$$\begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 2 \end{pmatrix} \begin{pmatrix} 2 & -1 & 0 \\ 1 & 3 & 1 \\ -2 & 0 & 4 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 0 & 1 \\ 3 & 2 \end{pmatrix}.$$

(2) Let $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

(a) Find a formula for A^n ($n \geq 1$) and use it to compute A^{10}. **[10 pts]**

(b) Calculate $f(A)$ for $f(x) = x^3 - 2x^2 + x + 3$. **[6 pts]**

3 Elimination matrices

Let $A = \begin{pmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{pmatrix}$ and $b = \begin{pmatrix} 2 \\ 12 \\ 2 \end{pmatrix}$.

(a) Find the elimination matrices E_{21} and E_{32} such that $E_{32}E_{21}A = U$ is upper triangular. Write down U. **[8 pts]**

(b) Compute the single matrix $E = E_{32}E_{21}$. **[4 pts]**

(c) Use elimination to solve $Ax = b$. **[6 pts]**

4 Inverse matrix

(1) Calculate the inverse of $Q = \begin{pmatrix} 1 & 3 & -2 \\ 1 & 1 & -1 \\ -2 & -1 & 2 \end{pmatrix}$. **[12 pts]**

(2) Use your answer to (1) to solve $Qx = (1, 2, 3)^T$. **[6 pts]**

5 $A = LU$

Let $A = \begin{pmatrix} 2 & 4 & -2 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{pmatrix}$.

(a) Factor $A = LU$. **[12 pts]**

(b) Solve $Ax = (2, 8, 10)^T$ by solving $Lc = b$ and then $Ux = c$. **[8 pts]**